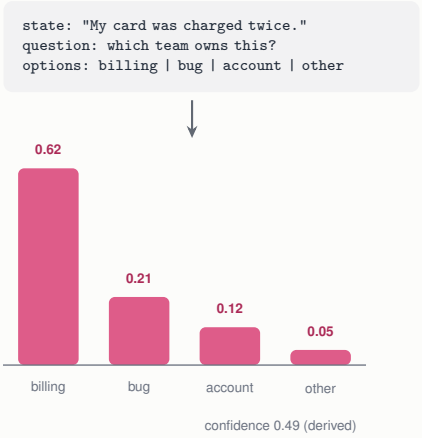
TUTORIAL · UNDERGRADUATE EDITION, 2026

Decision Models

Models that answer a multiple-choice question with probabilities instead of text. What a probability has to mean, how to act on it, and where it can mislead. Every term is defined as it appears; a glossary is at the end.



A decision model answers with probabilities

You supply the options. It returns how likely each one is. There is no text to read or parse.

a generative call: text comes back

```
"Which team owns this ticket: billing, bug,
 account or other? One word.
 Ticket: My card was charged twice."
-> "Billing."   (text: parse it, hope)
```

a decision call to TypeSafe's Jev†: numbers come back

```
"state": {"ticket": "My card was charged twice."},
"questions": {"team": {"type": "choice",
 "instructions": "Which team owns 'ticket'?",
"criteria": ["billing","bug","account","other"]}}
-> "choice": "billing", "confidence": 0.49,
  "probabilities": {"billing": 0.62, "bug": 0.21,
                    "account": 0.12, "other": 0.05}
```

What is different

The output is a list of probabilities over the options you supplied: nothing to parse, no format drift.

The probabilities are the product. Over many answers, "0.8" should be right about 80% of the time (slide 6).

INTUITION A generator writes you a paragraph. A decision model ticks a multiple-choice form and writes beside each box how sure it is.

NEW TERMS **Generative model**: a language model that writes text. **Classification**: picking one label from a fixed list, the **option set**. **Probability distribution**: one number per option, adding up to 1. **API call**: a request a program sends to a model over the network. † marks vendor documentation, blogs or commentary.

Why now: an agent makes many small judgments

Most steps in an agent are a yes/no, a pick-one or a rating. A text generator is a costly way to ask.



Why not ask a big model every time?

Cost and speed. A top large model lists $5 per million input tokens and takes seconds; a decision model lists $0.042 and about 100 milliseconds (ms)[†]: 120× cheaper per input token (derived).

No honest probability. Asked “how sure are you?”, chat models answer overconfidently (Xiong et al. 2023).

What it cannot do

Write, explain itself, reason through a hard case, or choose outside your list. Those go to a reasoning model or a person.

INTUITION A triage nurse sorts most patients in seconds and flags the ones who need the doctor.

NEW TERMS **Agent**: a program that lets a model take actions in a loop. **Guardrail**: a safety check on inputs or outputs. **Token**: a word piece, the unit models read and bill by. **Latency**: waiting time. **System 1** / **System 2** (Kahneman): fast snap judgment / slow reasoning; hence the vendor’s “System One”. **Escalate**: hand off to a person or stronger model.

Four question shapes: choice, score, noul, box

The shape you ask for fixes what comes back, and what the confidence number means.

Choice

asks one option from your list

returns a probability per option and a confidence

read it worked example below

Score

asks a level on an ordered (ordinal) scale

returns a probability per level and their average

read it 1.43† lies between levels 1 and 2

Noul

asks a yes/no question

returns one number p: the probability of yes

read it near 0.5 is "cannot tell"; certainty is $|2p - 1|$

Box

asks where something is in an image

returns a rectangle (bounding box)

read it a preview†, not yet for production use

Reading a Choice: the ticket on slide 2

$n = 4$ options; the largest probability is $p_{\max} = 0.62$.

confidence $= (p_{\max} - \frac{1}{n}) \,/\, (1 - \frac{1}{n}) = (0.62 - 0.25) \,/\, 0.75 = 0.49$

0 is an even split; 1 is all the probability on one option (docs' formula†).

INTUITION Four exam formats: multiple choice, rate on a scale, true/false, point to it. Each answer states how sure it is.

NEW TERMS **Argmax**: the option with the highest probability. **Confidence**: one number from 0 to 1 for how peaked the probabilities are. **Expected value**: the probability-weighted average (here, of the levels).

State and questions: facts in, judgments asked

One request carries the facts once and asks many small, independent questions about them.

one request: the facts, then two questions about them

```
"state": {
 "order": {"id": "A-104", "charges": [49, 49]},
 "refund_policy":
   "Duplicate charges are refunded."},
"questions": {
 "team": {"type": "choice", "criteria": [..]},
 "refund_ok": {"type": "noul", "instructions":
   "Is a refund due for ‘order‘ under
    ‘refund_policy‘?"}}
```

12.2×
cheaper in one call than in 13†

10.0×
faster: 0.27 s against 2.71 s†

vendor cookbook†: 13 questions about one long document, in one call or in 13 calls; 11 of the 13 answers identical

INTUITION Hand the examiner one case file and a sheet of short questions, not one essay prompt.

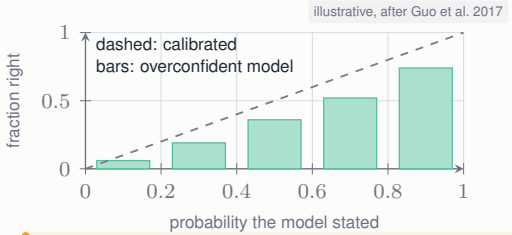
Rules for good questions

One judgment per question: broad ones hide several. **Rules stay in code:** compute dates, sums and counts first, then pass the result in as state.

NEW TERMS **State**: the facts of the case, sent once. **Question**: one typed judgment about the state. **Criteria**: the options (or level descriptions) for a question. **Batch**: several questions in one call; each is answered independently. **Deterministic rule**: a check that code can do exactly, such as “is it more than 30 days overdue?”

Calibration: what a probability has to mean

Of all the answers given with probability p, about a fraction p should turn out to be right.



INTUITION A forecaster who says “70% chance of rain” is calibrated if it rains on about 70% of the days she says so.

Reading the reliability diagram

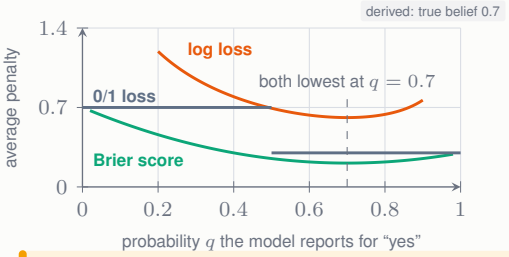
The plot. Group the answers by the probability the model stated. For each group, draw the fraction that were right. Calibrated answers land on the diagonal; bars below it mean the model claimed more than it delivered.

One number. Expected calibration error, $\mathrm{ECE} = \sum_b \frac{n_b}{N} |\mathrm{acc}_b - \mathrm{conf}_b|$: the average gap to the diagonal. n_b of the N answers are in group b; acc_b is its fraction right, conf_b its mean stated probability.

NEW TERMS **Calibration**: stated probabilities match how often the answers are right; a property of many answers, never of one. **Reliability diagram**: the plot above. **Bin**: one group of answers with similar probability. **Overconfident**: stating more than the hit rate. **Base rate**: how often something is true overall.

Proper scoring rules: grade the probability

Under a proper rule, the best expected score goes to the model that reports what it really believes.



INTUITION Mark only right or wrong, and bluffing pays. Also penalize confident misses, and honesty pays.

Why honesty can be the best policy

The setup. The model believes "yes" has probability 0.7 and reports q. Each curve is its average penalty.

Proper rules (Brier score, log loss) are lowest exactly at $q = 0.7$: reporting the true belief wins (Brier 1950; Gneiting and Raftery 2007).

Accuracy is not proper. Any $q > 0.5$ scores the same, so 0.99 costs no more than 0.7, and a guess beats "I don't know". Kalai et al. (2025): this is why models bluff ("hallucinate").

NEW TERMS **Scoring rule**: a formula grading a stated probability against what happened. **Proper**: the best average score comes from stating your true belief. **Brier score**: squared gap between probability and outcome (0 or 1). **Log loss** (cross-entropy): $-\log$ of the probability given to what happened. **0/1 loss**: 1 for a wrong top guess (accuracy).

Where calibration comes from, and where it goes

A plain language model's probabilities are fairly honest. Tuning it to please people spoils them.

four ways to get a probability out of a language model

Method	For	Against
Read logprobs of the option letters	one pass; cheapest	biased by option order and by the letters
Ask in words: "how sure are you?"	works on any chat model	overconfident, clustered (Xiong et al. 2023)
Sample k answers, count agreement	works for free-text answers	k times the cost; stability, not truth
Train a head: a decision model	trained in its answer format; one pass	fixed shapes; calibration claimed on vendor data

Calibration is easy to have and easy to lose

It starts out fine. A base language model is well calibrated on multiple-choice questions "in the right format" (Kadavath et al. 2022).

Post-training breaks it. GPT-4 report (2023): ECE 0.007 before, 0.074 after tuning on human preferences.

INTUITION Reward a student for sounding sure, and the student learns to sound sure.

NEW TERMS **Logit**: a model's raw score for a token; **logprob**: the log of its probability. **Base model**: trained only to predict text. **Post-training**: later tuning into an assistant, e.g. **RLHF**, reinforcement learning from human feedback (preferences). **Head**: a small output layer added to a model.

How a decision model is trained: what is known

The vendor gives a name and a goal. Published recipes with that goal reward a proper scoring rule.

three training signals, three kinds of model

Training	What is rewarded	Known weakness
RLHF	text that human raters prefer	sounds sure when wrong; calibration lost (slide 8)
RLVR	a final answer that a program verifies	overconfident when wrong (Mei et al. 2025)
RLCD†	calibrated probabilities over your options	undisclosed: no paper, no weights

INTUITION You get what you reward. Reward pleasing text, a checked answer, or honest odds, and you get three different models.

What is disclosed, and what is inferred

Disclosed†**.** A name, RLCD (reinforcement learning for calibrated decisions), and a goal: decisions and calibrated probabilities instead of text. No paper, weights or architecture.

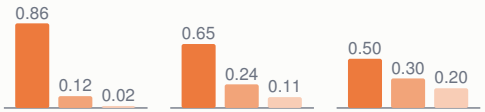
Inferred, not disclosed. The natural reward is a proper scoring rule. A published relative, RLCR (Damani et al. 2025), rewards correctness plus the Brier score of the stated confidence, and improves calibration at no cost in accuracy.

NEW TERMS **Reinforcement learning (RL)**: learning by trial, guided by a score called the **reward**. **Policy**: the model being trained, seen as a rule for what to output. **RLVR**: RL with verifiable rewards. **RLCD**: RL for calibrated decisions (the vendor's name). **RLCR**: RL with calibration rewards.

Fixing calibration afterwards: temperature scaling

One number, fitted on a few hundred labelled examples, repairs most of it. The top answer stays.

the same raw scores (2.0, 1.0, 0.2), three temperatures (derived)



$$p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

z_i: the raw score (logit) of option i
T: the temperature
p_i: the probability of option i

INTUITION A volume knob for confidence: louder or quieter, but the same answer stays on top.

What one number can and cannot fix

What it does. Divide every raw score by T before turning scores into probabilities. $T > 1$ flattens them, $T < 1$ sharpens them.

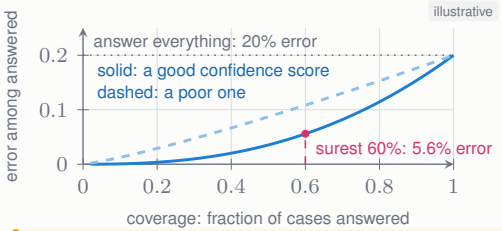
What it needs. A few hundred labelled examples from your own data; T is fitted on them (Guo et al. 2017).

Limits. Fitted on one kind of data, it is off on another. Decision models need it too: recalibrating Jev on 2,416 human labels cut the error 3.3× (Rafe and Das 2026).

NEW TERMS **Softmax**: the formula above; it turns raw scores into probabilities that add up to 1. **Temperature**: the divisor T inside it. **Held-out set**: labelled examples kept aside for fitting or testing. **Distribution shift**: new data unlike the data used for fitting. **Recalibration**: adjusting probabilities after training.

Answer when sure, hand off the rest

Act only above a probability threshold. Where the threshold goes depends on what each mistake costs.



INTUITION A new doctor treats the clear cases and refers the unclear ones. How unclear is too unclear depends on the stakes.

act when $p \geq p^* = c_{\mathrm{FP}} \,/\, (c_{\mathrm{FP}} + c_{\mathrm{FN}})$ (Elkan 2001)

Decision (illustrative costs)	c_{FP}	c_{FN}	p^*
flag a message for safety review	1	50	0.02
auto-approve a $49 refund	49	5	0.91
route a ticket to billing	2	2	0.50
re-run an expensive extraction	1	20	0.05

The threshold is a cost decision

c_{FP}: the cost of acting wrongly; c_{FN}: of wrongly holding back. One probability, four thresholds, all in your code.

NEW TERMS **Selective prediction**: answering only when confident enough. **Coverage**: the fraction of cases answered. **Selective risk**: the error rate among those. **Threshold** (p^*): the cut-off probability for acting. **False positive (FP)**: acting wrongly; **false negative (FN)**: wrongly not acting. **Deferral**: passing a case on.

Cascades: the cheap model decides who pays

A small calibrated model answers the easy cases and passes the hard ones to a big model or a person.

a two-stage cascade



77-way intent test[†]	right	per 1,000
Jev alone	81.0%	$0.11
cascade: hand off below 0.90	84.0%	$0.69
Claude Opus 5 alone	84.4%	$2.42

Banking77 test, measured by OpenRouter[†]. FrugalGPT (Chen et al. 2023): a cascade matched GPT-4 at up to 98% lower cost.

Sets with a guarantee: conformal prediction

Instead of one answer, return every option above a cut-off fitted on labelled examples, so that the set holds the right answer a chosen fraction of the time (written $1 - \alpha$, say 90%). One option in the set: act. Several: ask. The guarantee holds while new cases resemble those examples (Angelopoulos and Bates 2021).

INTUITION A help desk: the front line settles the routine calls and escalates the rest to a specialist.

NEW TERMS **Cascade**: models in a chain, cheapest first; a case moves on only when the current model is unsure. **Router**: the rule that picks who handles a case. **Conformal prediction**: turning probabilities into a set of options with a **coverage guarantee**, valid while new cases resemble the fitting cases (**exchangeable**).

The evidence so far, October 2026

Independent tests: close to large models at picking the answer; calibration depends on the task.

Jev, released September 15, 2026†: $0.042 per million input tokens, 78–127 ms median, 956M requests in one week

Test (size)	Jev	Compared with	Measured by
Film-review sentiment (25,000)	96.5% right	a small fine-tuned classifier: “similar”	Raschka†
Approve an agent’s tool call (100)	93% zero-shot	Claude Sonnet 5: 98%; majority guess: 79%	Archestra†
31 yes/no alignment-failure sets	AUROC 0.886 (median)	median ECE per set 0.168; perfect calibration would give 0.074	Guo et al. 2026
Complaint outcomes (2,400)	45%; at best 55%	a word-count model (XGBoost) did better	Anaconda†

The pattern

Near large models at telling right from wrong, for a few percent of the cost and a tenth of the wait. How sure to be is where it slips.

INTUITION A sharp sorter is not automatically a good judge of its own certainty.

NEW TERMS **Discrimination**: telling right answers from wrong ones; a different property from calibration. **AUROC** (area under the receiver operating characteristic curve): the chance that a right answer gets higher confidence than a wrong one. **Zero-shot**: no worked examples given. **Alignment failure**: an AI system acting against its rules.

Limits: where a decision model is rough

It reads literally, leans toward the first option, and can be steered by text planted in a request.

three measurements, on a scale from 0 to 1

Wording. The same question reworded: score 0.708 to 0.946

Planted text. Correct decisions flipped: 61% to 73%

No base rate. “Which side did a fair coin land on?”



Black dot: median 0.874. TypedBench, 500 support tickets; JevOut (Xu et al. 2026), four decision systems; community tests†.

INTUITION A fast, literal clerk: follows the form to the letter and believes what is in the file.

What goes wrong, and what helps

The vendor’s own list†. Takes criteria at their word; unreliable at counting, arithmetic and dates; leans toward the first option; irrelevant context lowers accuracy; cannot say why.

A probability is about the question, not the world: nothing in a request says the coin is fair.

What helps. Shuffle the options and average; list “none of these”; keep the state short; put hard rules in code, first.

NEW TERMS **Position bias**: favouring an option because of where it is listed. **Paraphrase**: the same question in other words. **Context rot**: accuracy lost to irrelevant text in the request. **Adversarial input**: text written to steer the model. **Deny-list**: actions that code always blocks.

Summary: what to take from this lecture

The science is old, the economics are new, and the evidence is young.

What to remember

1. A decision model returns probabilities over options you supply. The probability is the product.

2. Four shapes: choice, score, noul (yes/no), box. Confidence says how peaked the probabilities are; it is no guarantee.

3. Calibration: stated probabilities match hit rates over many answers. Proper scoring rules reward it; tuning on preferences erodes it.

4. A threshold comes from what each mistake costs, and it lives in your code.

5. Cascade: the cheap model first; a large model or a person for the unsure cases.

6. Recalibrate and test on your own labelled data; treat the state as untrusted.

Still open, October 2026

Is RLCD more than cross-entropy plus a Brier term? There is no paper.

Can calibration survive planted text? Flip rates of 12–73% in two studies say not yet.

What is the right test? Accuracy leaderboards reward guessing.

Who owns the decision? Thresholds, deny-lists and logs are where “code decides” is kept or broken.

You can build one too

Read a stock model’s logprobs, add a head to a frozen model, or fine-tune a small classifier. Open-weight versions (downloadable parameters) exist[†].

Glossary, part 1

Every term and abbreviation from the slides, in alphabetical order. † marks a vendor's own term.

0/1 loss 1 for a wrong top guess, else 0.
Adversarial input text written to steer the model.
Agent a program that lets a model act in a loop.
Alignment failure an AI system acting against its rules.
API application programming interface; a call is a request to a model.
Argmax the option with the highest probability.
AUROC area under the receiver operating characteristic curve: how well confidence ranks right answers above wrong ones.
Base model a model trained only to predict text.
Base rate how often something is true overall.
Batch several questions in one call.
Bin answers grouped by similar probability.
Box an answer that is a rectangle (bounding box).
Brier score squared gap between probability and outcome.
Calibration stated probabilities match hit rates.
Cascade cheapest model first; unsure cases move on.
Choice an answer that is one option from a list.
Classification picking one label from a fixed list.
Confidence how peaked the probabilities are, 0 to 1.
Conformal prediction option sets with a coverage guarantee.
Context rot accuracy lost to irrelevant text.
Coverage the fraction of cases answered.
Criteria the options or levels of a question.
Deferral, escalation passing a case to a person or a stronger model.
Deny-list actions that code always blocks.
Deterministic rule a check code can do exactly.
Discrimination telling right answers from wrong.
Distribution shift new data unlike the fitting data.
ECE expected calibration error: average gap between stated probability and hit rate.
Exchangeable new cases resemble the fitting cases.
Expected value the probability-weighted average.
False positive / negative (FP / FN) acting wrongly / wrongly not acting.

Glossary, part 2

Generative model to Zero-shot.

Generative model a model that writes text.
Guardrail a safety check on inputs or outputs.
Head a small output layer added to a model.
Held-out set labelled examples kept aside.
Latency waiting time; ms = milliseconds.
Log loss cross-entropy: minus the log of the probability given to what happened.
Logit, logprob raw score for a token; log of its probability.
Noul a yes/no answer given as one probability.
Option set the fixed list of labels to choose from.
Overconfident stating more than the hit rate.
Paraphrase the same question in other words.
Policy the model being trained, as a rule for what to output.
Position bias favouring an option for where it is listed.
Post-training tuning a base model into an assistant.
Probability distribution one number per option, summing to 1.
Proper scoring rule grading that is best satisfied by stating your true belief.
Recalibration adjusting probabilities after training.
Reliability diagram hit rate plotted against stated probability.
RL reinforcement learning: learning by trial, guided by a reward.
RLCD† RL for calibrated decisions.
RLCR RL with calibration rewards.
RLHF RL from human feedback.
RLVR RL with verifiable rewards.
Router the rule that picks who handles a case.
Score an answer on ordered (ordinal) levels.
Selective prediction answering only when confident.
Selective risk the error rate among cases answered.
Softmax turns raw scores into probabilities summing to 1.
State, question the facts, sent once; one typed judgment about them.
System 1 / **System 2** fast snap judgment / slow reasoning.
Temperature the divisor T inside the softmax.
Threshold (p^*) the cut-off probability for acting.
Token a word piece; the unit of reading and billing.
Zero-shot no worked examples given.

Sources and further reading

Papers first, then vendor and community material; † marks vendor documentation, blogs and commentary.

Papers cited on the slides

Calibration and scoring. Brier 1950; Gneiting & Raftery 2007; Guo et al. 2017 (temperature scaling); Kadavath et al. 2022; OpenAI 2023 (GPT-4 technical report); Xiong et al. 2023; Kalai et al. 2025.

Training for calibration. Damani et al. 2025 (RLCR); Mei et al. 2025.

Deciding. Elkan 2001 (cost-sensitive thresholds); Chen et al. 2023 (FrugalGPT); Angelopoulos & Bates 2021 (conformal prediction).

Fast and slow thinking. Kahneman 2011.

Decision models, 2026. Rafe & Das 2026; Guo et al. 2026; Xu et al. 2026 (JevOut); Hu et al. 2026 (JevAdvBench); Sharma et al. 2026 (TypedBench, arXiv 2610.11392).

Vendor and community material†

TypeSafe documentation (docs.typesafe.ai, read Oct 21, 2026): System One, state, the primitives (choice, score, noul, box), confidence, model jaggedness for Jev 1.13, and the parallel-questions cookbook.

OpenRouter: Decisions guide and rankings; the Jev against Claude Opus 5 classification post; the Jev-verified cascade cookbook.

Artificial Analysis Decision Index (latency).

Independent tests: Raschka, “Classifier history and Jev”; Archestra, “We tested Jev on 100 real agent calls”; Anaconda, “Decision time”; community calibration tests.

Curves labelled *illustrative* are sketches, not data. Numbers labelled *derived* were computed from quoted ones. Everything about Jev and the October 2026 landscape is carried over from the long deck as stated there.

The full reference list (about 90 sources, with arXiv identifiers) is in the original 32-slide deck and its paper kit.